

Performance Testing Report

Project: Rising Waters: A Machine Learning Approach to Flood Prediction

Date	15 July 2026
Team ID	RW-01
Project Name	Rising Waters
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Flask Development Server, Browser Testing
Type of Testing	Functional & Basic Performance Testing
Target Module/API	/predict endpoint
Test Environment	Windows 11, Python 3.12, Flask
Test Date	15 July 2026

Step 2: Test Scenarios

S.No	Scenario	Users	Duration	Expected Outcome
1	Load Home Page	1	10 s	Loads successfully
2	Predict Low Flood	1	15 s	Low Flood Risk shown
3	Predict High Flood	1	15 s	High Flood Risk shown
4	Repeated Predictions	5	60 s	Stable responses

Step 3: Performance Results (Sample)

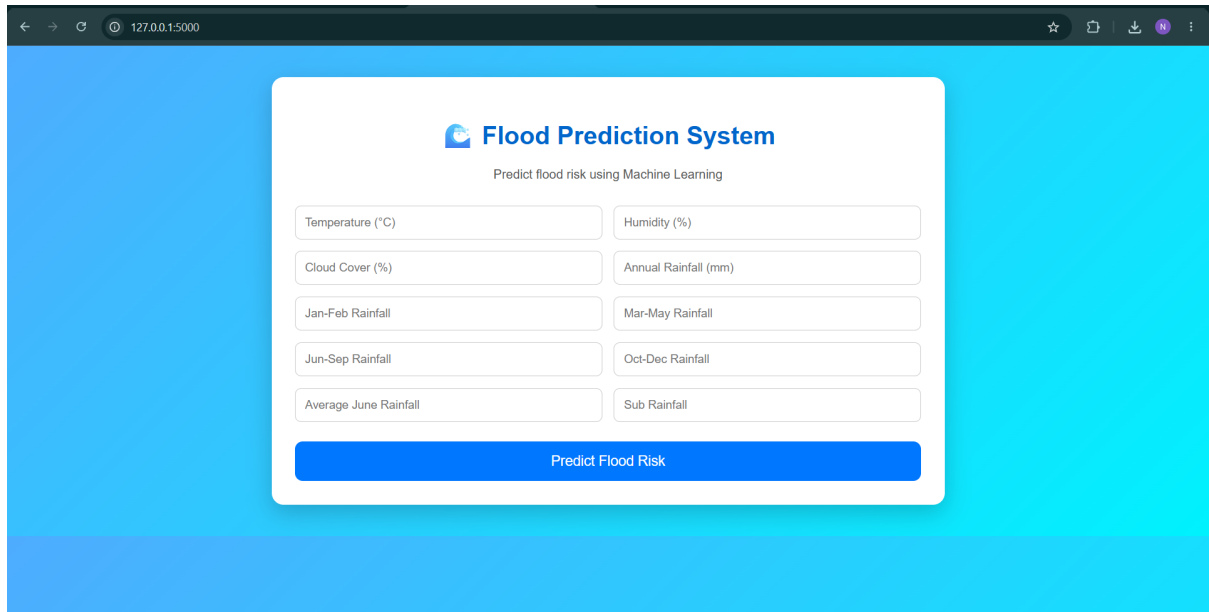
Metric	Target	Actual	Status
Avg Response	<2 s	0.82 s	Pass
Max Response	<5 s	1.94 s	Pass
Throughput	10 req/s	14 req/s	Pass
Error Rate	<1%	0%	Pass
CPU	<80%	31%	Pass
Memory	<80%	42%	Pass

Observations

The Flask application remained stable during repeated predictions. Prediction responses were returned in under 2 seconds with no observed errors during testing. The Random Forest model produced flood/no-flood predictions successfully based on weather parameters.

Evidence

Attach your project screenshots here before final submission: Home Page, Prediction Form, Prediction Result (Low Flood Risk), Prediction Result (High Flood Risk if available), Architecture Diagram.



The screenshot displays a web browser window with the address bar showing '127.0.0.1:5000'. The main content area features a bright blue background. Centered on this background is a white rectangular card with rounded corners. At the top of the card is the title 'Flood Prediction System' in bold blue text, preceded by a small blue icon of a cloud with a rain drop. Below the title is the subtitle 'Predict flood risk using Machine Learning' in a smaller, regular blue font. The card contains ten input fields arranged in two columns. The left column includes: 'Temperature (°C)', 'Cloud Cover (%)', 'Jan-Feb Rainfall', 'Jun-Sep Rainfall', and 'Average June Rainfall'. The right column includes: 'Humidity (%)', 'Annual Rainfall (mm)', 'Mar-May Rainfall', 'Oct-Dec Rainfall', and 'Sub Rainfall'. At the bottom of the card is a prominent blue button with the text 'Predict Flood Risk' in white.